EEE223 CIRCUIT THEORY I LABORATORY

EXPERIMENT-1 / MEASUREMENT OF VOLTAGE, CURRET, AND RESISTANCE

1. INTRODUCTION

The purpose of this experiment is measurement of current, voltage, and resistive quantities by use of an AVO meter and the experimental verification of Kirchhoff's current and voltage Laws.

2. PRELIMINARY WORK

2.1

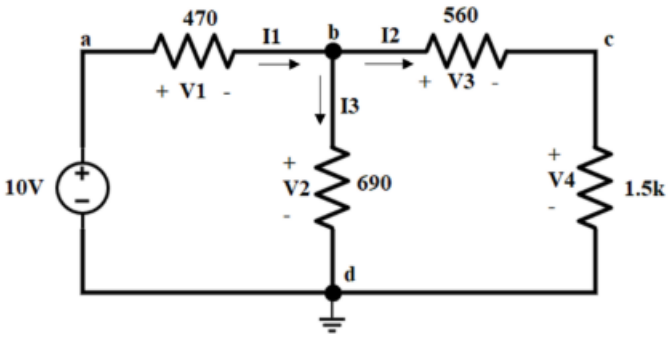


Figure 2.1

I1(mA)	I2(mA)	I3(mA)	Rad(Ω)	V1(V)	V2(V)	V3(V)	V4(V)
10,13	2,54	7,59	986.87	4,76	5,24	1,42	3,82

Table 2.1

2.2

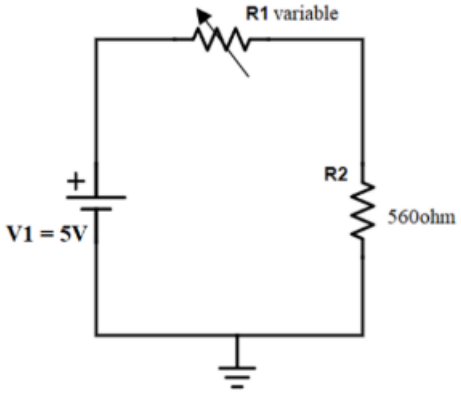
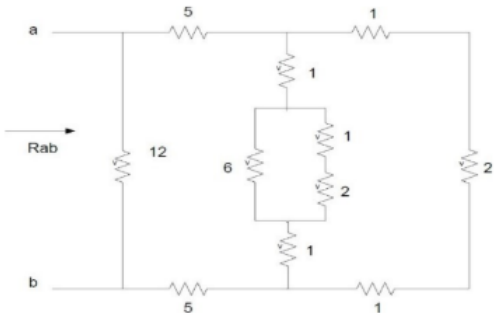


Figure 2.2

	Variable resistor (4k Ω)	Variable resistor (8k Ω)
Current (mA)	1,10	0,58
R2 Voltage (V)	0,6	0,33

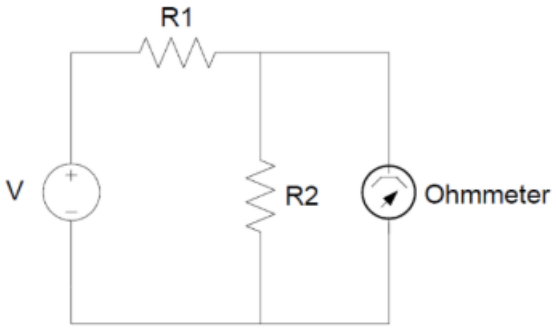
Table 2.2

2.3



Rab = 6 Ω

2.4



This type of connection does not the correct value because an ohmmeter should never be connected to an active circuit; the external voltage source (V) interferes with the measurement and may damage the meter. Additionally, R2 is still connected in parallel with the rest of the circuit (R1 and the source), which alters the measured equivalent resistance. To measure R2 correctly, the power source must be disconnected and at least one terminal of R2 isolated from the circuit.

2.5

Resistor Values	Color Codes			
	A	B	C	D
100Ω±10%	Brown	Black	Brown	Silver
120Ω ±5%	Brown	Red	Brown	Gold
220Ω ±20%	Red	Red	Brown	-
330Ω ±10%	Orange	Orange	Brown	Silver

Table 2.5

3. SIMULATION WORKS

3.1

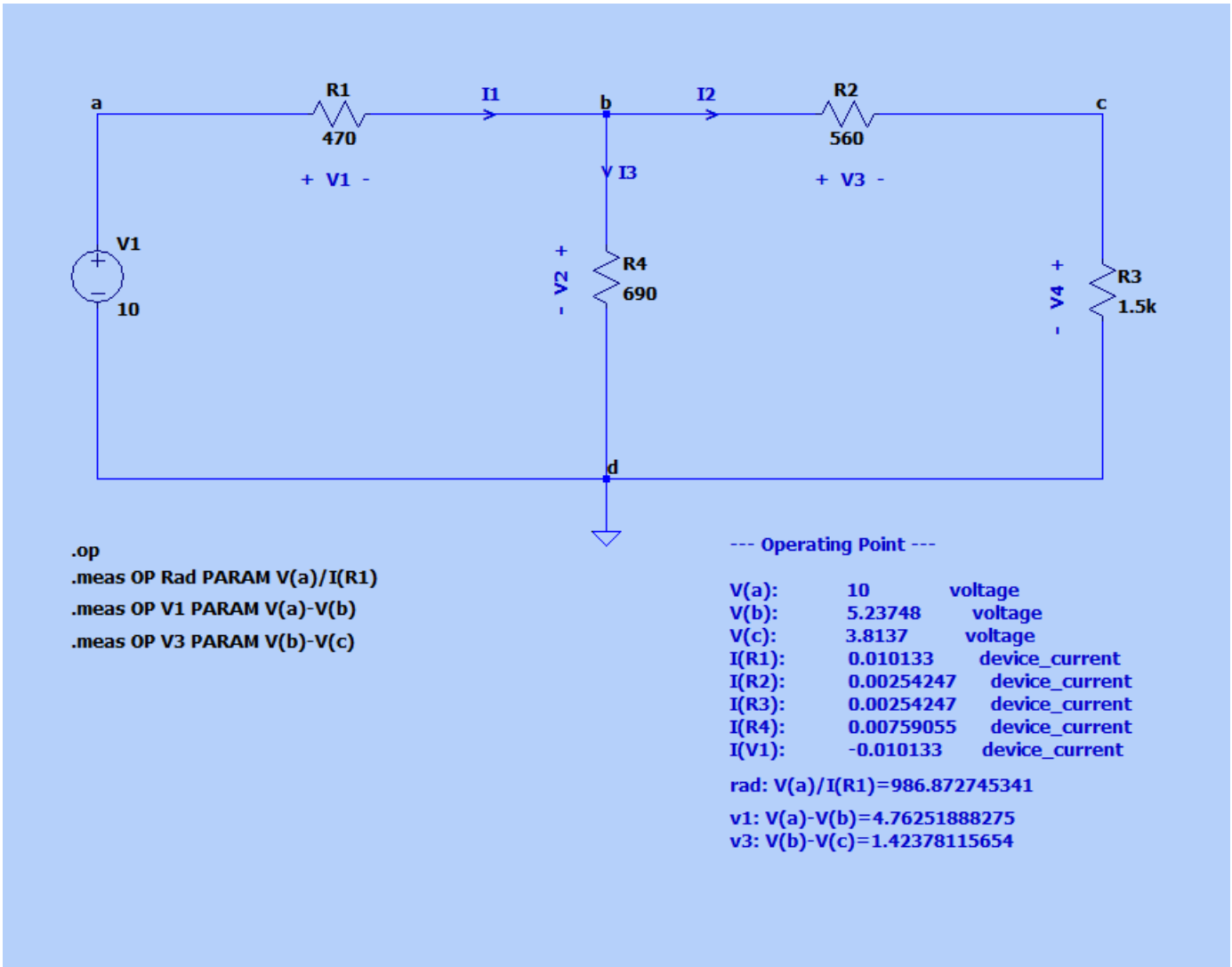


Figure 3.1

I1(mA)	I2(mA)	I3(mA)	Rad(Ω)	V1(V)	V2(V)	V3(V)	V4(V)
10,133	2,542	7,590	986.872	4,762	5,237	1,423	3,813

Table 3.1

3.2

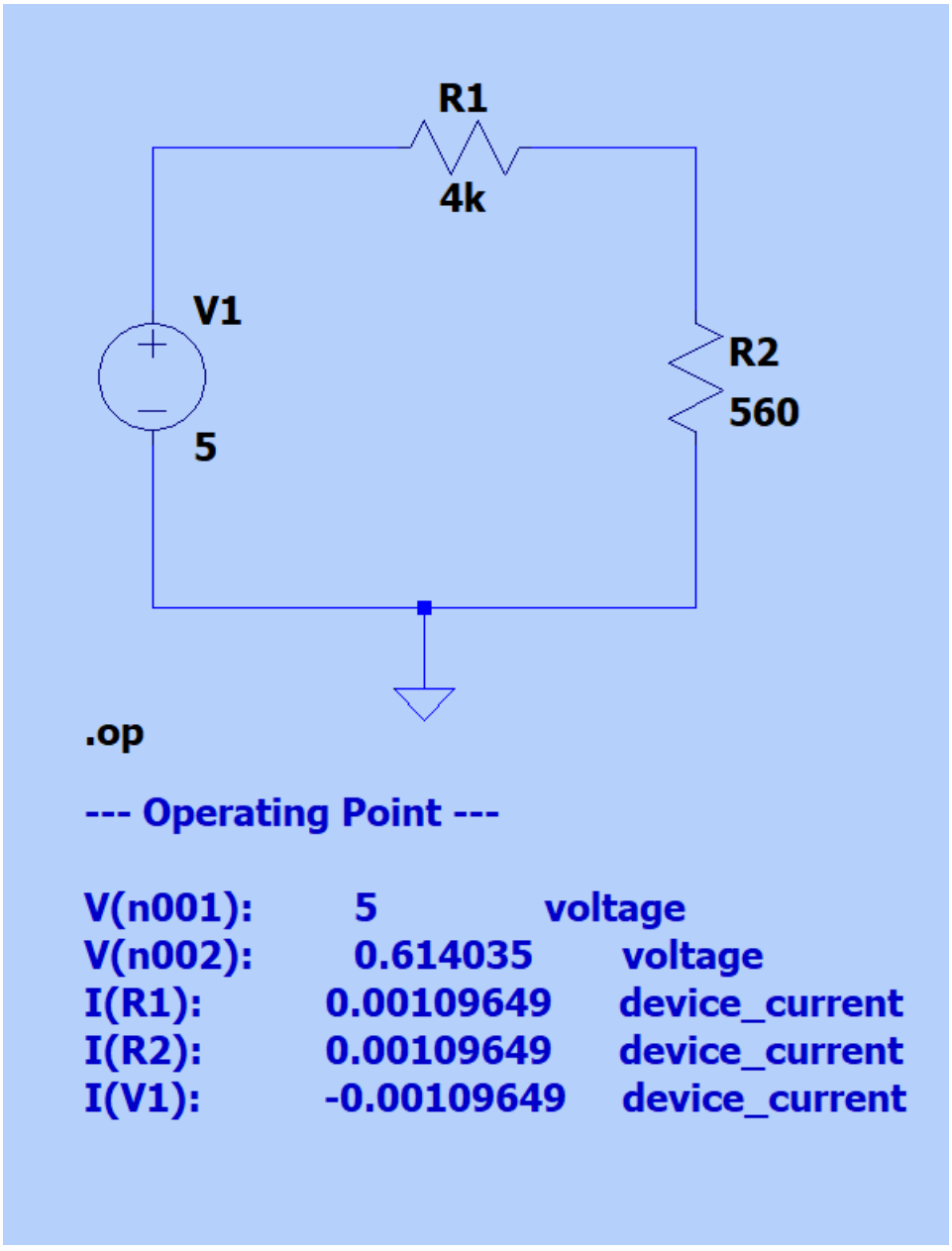


Figure 3.2(a)

	Variable resistor (4k Ω)
Current (mA)	1,096
R2 Voltage (V)	0,614

Table 3.2(a)

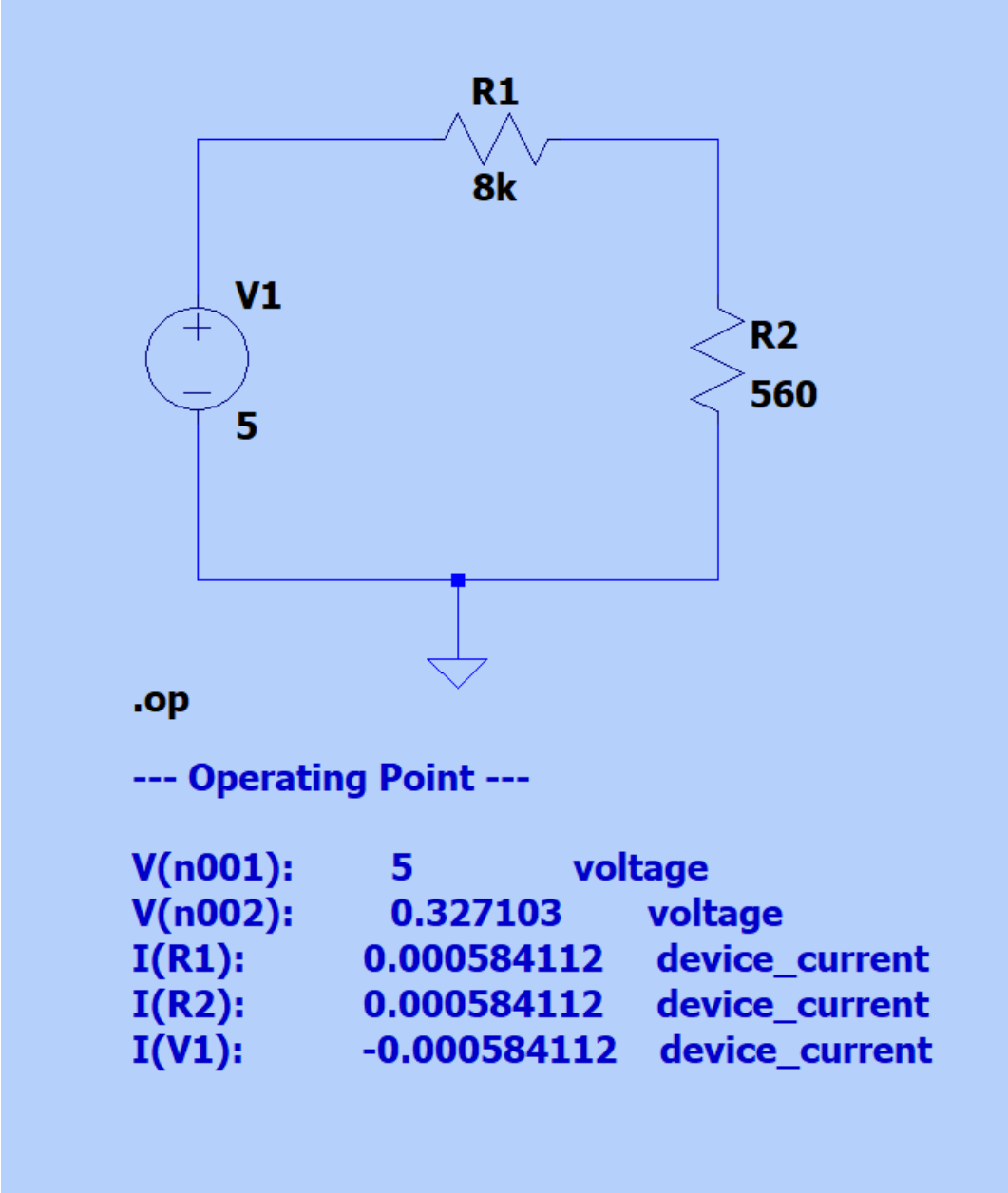


Figure 3.2(b)

	Variable resistor (8k Ω)
Current (mA)	0,584
R2 Voltage (V)	0,327

Table 3.2(b)

4. EXPERIMENT RESULTS

4.1

I1(mA)	I2(mA)	I3(mA)	V1(V)	V2(V)	V3(V)	V4(V)	Rad(Ω)	R1(Ω)	R2(Ω)	R3(Ω)	R4(Ω)
10,14	**2,53**	**7,61**	**4,75**	**5,21**	**1,5**	**3,8**	**977**	**465**	**676**	**548**	**1487**

Table 4.1

***this values is experimental result for the figure 2.1*

***the directions of current and voltage have been taken as shown in the figure 2.1, and the values are indicated as positive.*

4.2

	Variable resistor (4k Ω)	Variable resistor (8k Ω)
Current (mA)	1,09	0,5
R2 Voltage (V)	0,6	0,277

Table 4.2

***this values is experimental result for the figure 2.2*

*** values are indicated as positive, based on the assumption that the current flows from the positive terminal of the source.*

4.3

R1 value (Ω)	Calculated Current (mA)	Calculated Voltage (V)
	2,5	1,4
	Measured Current (mA)	**Measured Voltage (V)**
	2,56	1,42

Table 4.3

***this values for the figure 2.2 when the current will be 2,5 mA*

5. CONCLUSIONS

In this experiment, the operational principles of an AVO meter (multimeter) in DC voltage, current, and resistance measurement modes were investigated, and fundamental circuit laws (Ohm's Law, KCL, and KVL) were verified across theoretical calculations, computer simulations, and experimental measurements.

In the first part of the experiment (Figure 2.1), node voltages and branch currents were analyzed. Theoretical calculations yielded an equivalent input resistance of Rad, a total current of I1, and branch currents of I2 and I3. Kirchhoff's Current Law at node b and Kirchhoff's Voltage Law around the loops were confirmed by the simulation and measured experimental data within acceptable measurement error tolerances. Minor discrepancies between theoretical and measured values stem from real component tolerances (resistor color code tolerances of (5% to 10%), internal resistances of the measurement probes and ammeter, and the non-ideal characteristics of the DC power supply.

The proper use of measurement instruments was also verified. Connecting the ammeter in series ensures that the branch current flows directly through the instrument, taking advantage of its low internal resistance to prevent circuit loading. Conversely, connecting the voltmeter in parallel across components allows accurate voltage drops to be recorded without drawing significant current due to its high input impedance. For resistance measurements, isolating components from the active circuit proved necessary; taking in-circuit measurements while the source is connected results in false readings and risks meter damage due to parallel sneak paths and active potential differences.

In the variable resistance section (Figure 2.2), the inversely proportional relationship between total circuit resistance and current was demonstrated. For potentiometer settings of 4kΩ and 8kΩ, the measured current decreased from 1,10 mA to 0,58 mA, resulting in corresponding drops across R2(560 Ω). To achieve a target circuit current of 2,5 mA with a 5 V source, the total resistance required is:

Rtotal = 5 V / 2,5 mA = 2000 Ω

Subtracting R2 = 560 Ω yields a required potentiometer value of:

R1 = 2000 – 560 = 1440 Ω

Setting R1 = 1,44 kΩ yielded an R2 voltage of VR2 = 2,5 mA * 560 Ω = 1,4 V, confirming the voltage divider and current regulation behavior of a potentiometer configured as a rheostat.

Overall, the experimental results demonstrate close agreement with both theoretical derivations and simulation models, confirming the validity of basic circuit analysis techniques and instrument operation procedures.